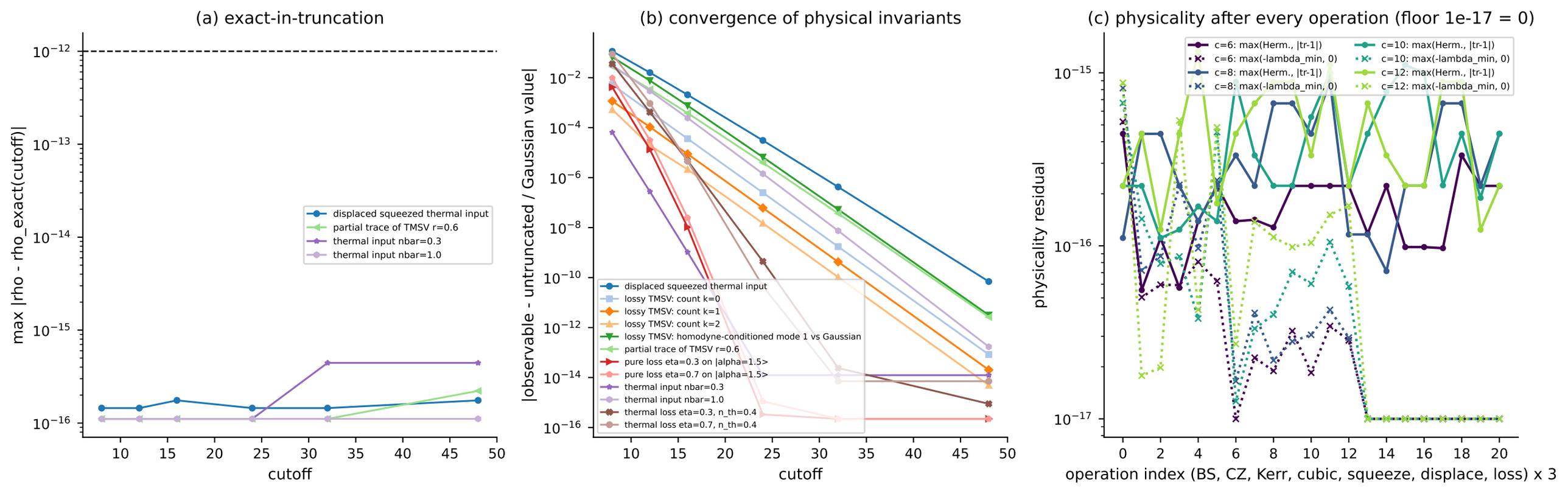
(a) exact-in-truncation
max |rho - rho_exact(cutoff)|
cutoff
displaced squeezed thermal input
partial trace of TMSV r=0.6
thermal input nbar=0.3
thermal input nbar=1.0
(b) convergence of physical invariants
|observable - untruncated / Gaussian value|
cutoff
displaced squeezed thermal input
lossy TMSV: count k=0
lossy TMSV: count k=1
lossy TMSV: count k=2
lossy TMSV: homodyne-conditioned mode 1 vs Gaussian
partial trace of TMSV r=0.6
pure loss eta=0.3 on |alpha=1.5>
pure loss eta=0.7 on |alpha=1.5>
thermal input nbar=0.3
thermal input nbar=1.0
thermal loss eta=0.3, n_th=0.4
thermal loss eta=0.7, n_th=0.4
(c) physicality after every operation (floor 1e-17 = 0)
physicality residual
operation index (BS, CZ, Kerr, cubic, squeeze, displace, loss) x 3
c=6: max(Herm., |tr-1|)
c=6: max(-lambda_min, 0)
c=8: max(Herm., |tr-1|)
c=8: max(-lambda_min, 0)
c=10: max(Herm., |tr-1|)
c=10: max(-lambda_min, 0)
c=12: max(Herm., |tr-1|)
c=12: max(-lambda_min, 0)